

Your Code

```
1 accounts = {}
2
3 def create_account(acc_no, name, balance):
4     accounts[acc_no] = {'name': name, 'balance': balance, 'txns': []}
5
6 def deposit(acc_no, amount):
7     if acc_no in accounts:
8         accounts[acc_no]['balance'] += amount
9         accounts[acc_no]['txns'].append(('Deposit', amount))
10
11 def withdraw(acc_no, amount):
12     if acc_no in accounts:
13         if accounts[acc_no]['balance'] >= amount:
14             accounts[acc_no]['balance'] -= amount
15             accounts[acc_no]['txns'].append(('Withdraw', amount))
16         else:
17             print('Insufficient balance')
18
19 def mini_statement(acc_no):
20     for t in accounts[acc_no]['txns'][-5:]:
21         print(t)
22
23 create_account(1001, 'Ravi', 10000)
24 deposit(1001, 5000)
25 withdraw(1001, 20000)
26 withdraw(1001, 3000)
27 mini_statement(1001)
28 print('Balance:', accounts[1001]['balance'])
```

Input

stdin input

Output

```
Insufficient balance
('Deposit', 5000)
('Withdraw', 3000)
Balance: 12000
```



Q2

Student Grade Book
Application
10 marks



2/2 answered

- (b) Computes total, average, and grade (O: ≥ 90 , A+: ≥ 80 , A: ≥ 70 , B+: ≥ 60 , B: ≥ 50 , F: < 50) for each student.
(c) Finds the class topper (highest average) and the subject with the highest overall mean.
(d) Displays a formatted result table sorted by average in descending order.
Use dictionary comprehension and list operations. Minimum 5 students.
Write complete, syntactically correct Python programs using lists, tuples, dictionaries, and sets with proper indentation.

Your Code

```
1 students={"A": [90, 80, 70, 60, 50], "B": [95, 92, 88, 90, 85], "C": [60, 55, 65, 70, 68]}
2 subs=["S1", "S2", "S3", "S4", "S5"]
3 grade=lambda a: "O" if a >= 90 else "A+" if a >= 80 else "A" if a >= 70 else "B" if a >= 60 else "F"
4
5 res={n: {'t': sum(m), 'a': sum(m)/5, 'g': grade(sum(m)/5)} for n, m in students.items()}
6 top=max(res, key=lambda x: res[x]['a'])
7 means=[sum(students[s][i] for s in students)/len(students) for i in range(5)]
8 best=means.index(max(means))
9
10 print("Name Tot Avg Grade")
11 for n in sorted(res, key=lambda x: res[x]['a'], reverse=True):
12     print(n, res[n]['t'], f"{res[n]['a']:.2f}", res[n]['g'])
13 print("Topper:", top)
14 print("Best Subject:", subs[best])
```



Run



Save

Input

stdin input

Output

```
Name Tot Avg Grade
B 450 90.00 O
E 373 74.60 A
A 350 70.00 A
C 318 63.60 B+
D 225 45.00 F
```